

When Equal Rules Become Unfair: AI, Finance, and the Hidden Advantage

Artificial Intelligence (AI) is an essential tool used worldwide. The most common type of AI in daily life is Large Language Models (LLMs). However, AI can do more than chat or answer questions. It can also make deep analyses based on numbers. For example, AI bots can perform financial analysis and make forecasts. This potential may reduce the human factor in financial markets. It may also reduce emotions, intuition, and personal judgment, turning finance into mere mathematics.

To learn more and improve my programming skills, I created a stock-forecasting AI bot. I did research and experiments to find the best parameters for my bot. I tried different scenarios by changing parameters and testing ideas so that the model would neither overfit nor underfit. I used Long Short-Term Memory (LSTM), a model used for numerical data such as finance or climatology. This bot makes 10-day forecasts on financial assets whose data is sourced from Yahoo Finance through its API. I also created a website to share my research project: hamdiai.com.

On the website, I show the accuracy rates for different stocks. These accuracy scores are determined by flexible rules, mostly focusing on the direction of price movement without ignoring the amount of change. These rules were shaped during my trials on Investopedia Simulator. The test data includes the days between January 1 2025, and 17 March 2026. For some well-known stocks, the accuracy rates were between 70% and 80%. This result shows the potential power of AI in the stock market. Still, when this project was made by a 17-year-old student with limited resources, it raises an important question: What can huge banks or fund companies create with stronger computers, larger data sources, and professional software teams?

Unfortunately, we do not fully know the calculations behind the system because we only provide numerical data to the model. Then the model finds and learns sequences and repetitive structures inside these number lists. This is difficult for humans to understand because LSTM models perform many calculations on time-series data. At this point, we can define financial AI bots as “black box models.” A black box model is a system where the user gives input data and receives an output, but the user does not fully know how the system reached that result.

The advantage of these models is that they process data fast. However, they also create a controversial ethical question: In an automated market where calculations are hidden inside a black box, to what extent can we still consider the human factor? The black box system does not involve human-based intuition and does not allow humans to recognize the mathematics behind the decision. In my project, I could see that some outputs were accurate, but I could not clearly

see all the calculations. Even if we could see them as equations, they would probably be too complex for humans to understand. This helped me understand the notion of the black box through first-person experience.

As mentioned before, I created a fairly accurate AI bot with limited resources and knowledge. This means that if a group with better computers, education, and experience creates a model, it will probably be more accurate than mine. When banks train AI bots, they can make profitable investments with less risk. As a result, power holders may earn more money and gain more influence over time. If rich people create better AI models, they will have a stronger chance to profit, while ordinary people may become weaker in the same market. Therefore, the gap between rich and poor can widen.

AI forecasts can create inequality of opportunity. People who think that everyone has access to artificial intelligence may fall into the illusion of equal opportunity. There will likely be more powerful models working in the background. Banks and companies can use this potential and have a better chance to win against ordinary people who try to research companies and analyze them by themselves. Humans will have difficulty competing with AI in finance if there are no ethical limits.

To sum up, my experience showed me that finance is becoming a battlefield of artificial intelligence algorithms. The human factor is slowly being removed from finance. AI is gradually hiding finance inside a black box. Financial markets should not become places where wealthy people make money from middle-class people with an unfair advantage. Markets should be fair for everyone, with real opportunities. If we do not guide these AI models with ethical principles, transparency, and responsibility, we may slowly lose control.

Başvururken, formu doldururken verdiğimiz cevaplar

Brief Description of Your Submission

This essay explores the ethical problems and the illusion of equal opportunity created by the integration of Artificial Intelligence in financial markets. Based on a machine learning model I developed, this article discusses how finance has become an unfair algorithmic battleground.

Reflection: What new insight or skill did you gain about AI Ethics, Responsible AI, AI Fairness, or AI for Social-Good while creating this piece?*

By working on my AI Finance bot, and writing this article I have learned that black box models can be dangerous in some ways. This threat is caused by its uncertainty. Since they are called black box models we can't know the calculations or the operations that it makes come up with

an output. Making unknown operations is problematic because we can't supervise. Being unsupervised for machines can create trouble such as causing people to make bad investment which can impact users' life. Some people who believes everything AI gives may not be able to understand the risk.